

RxSense — AI Workflow Cost Breakdown

RxSense — AI Workflow Cost Breakdown

Exchange rate: \$1 USD = ₹110 BDT
All estimates are per-call / per-session unless noted.

Pricing Reference

Model	Input (per 1M tokens)	Output (per 1M tokens)
Claude Sonnet 4.6	\$3.00	\$15.00
GPT-4.1	\$2.00	\$8.00
GPT-4o	\$2.50	\$10.00
GPT-4o-mini	\$0.15	\$0.60
text-embedding-3-small	\$0.02	—
PrescriptoAI OCR	\$0.12 flat	—
Modal L4 GPU	\$0.000533 / sec	—

Per-Workflow Breakdown

1. Prescription Upload — PrescriptoAI OCR

File	backend/src/controllers/prescriptionController.js
Model	PrescriptoAI (third-party OCR)
Type	Flat rate
Cost (USD)	\$0.12 / scan
Cost (BDT)	₹13.20 / scan

2. Prescription Upload — Modal MedGemma GPU (Fallback)

File	python-service/modal_app.py
Model	MedGemma on NVIDIA L4 GPU (Modal serverless)

GPU rate	\$0.000533 / sec
Est. processing	~30 seconds
Cost (USD)	~\$0.016 / scan
Cost (BDT)	~৳1.76 / scan

⚠️ Only invoked if PrescriptoAI is unreachable.

3. Dosage Safety Pass — GPT-4.1

File	backend/src/controllers/prescriptionController.js
Model	GPT-4.1 (DOSAGE_MODEL)
max_tokens	1,024
Input	~1,000 tokens (extracted med list + system prompt)
Output	~800 tokens (dosage annotations)

Line item	Calculation	Cost
Input	$(1,000 \div 1M) \times \$2.00$	\$0.0020
Output	$(800 \div 1M) \times \$8.00$	\$0.0064
Total (USD)		~\$0.0084 / scan
Total (BDT)		~৳0.92 / scan

4. Report Analysis — Lab / Imaging

File	backend/src/utils/reportAnalysisUtils.js
Model	GPT-4.1 (OPENAI_REPORT_MODEL)
max_tokens	8,192
Input	~1,700 tokens (system + extraction prompt + image tokens)
Output	~3,000 tokens (full JSON extraction)

Line item	Calculation	Cost
Input	$(1,700 \div 1M) \times \$2.00$	\$0.0034

Line item	Calculation	Cost
Output	$(3,000 \div 1M) \times \$8.00$	\$0.0240
Total (USD)		~\$0.027 / report
Total (BDT)		~৳3.00 / report

5. Report Chat — Q&A on Uploaded Report

File	<code>backend/src/controllers/reportChatController.js</code>
Model	GPT-4o-mini
max_tokens	500
Input	~1,000 tokens (system + report context + user message)
Output	~500 tokens

Line item	Calculation	Cost
Input	$(1,000 \div 1M) \times \$0.15$	\$0.00015
Output	$(500 \div 1M) \times \$0.60$	\$0.00030
Total (USD)		~\$0.00045 / message
Total (BDT)		~৳0.05 / message

6. Prescription Chat — Q&A on Scanned Prescription

File	<code>backend/src/controllers/chatController.js</code>
Model	GPT-4o-mini
max_tokens	500
Input	~800 tokens (system + prescription context + user message)
Output	~500 tokens

Line item	Calculation	Cost
Input	$(800 \div 1M) \times \$0.15$	\$0.00012
Output	$(500 \div 1M) \times \$0.60$	\$0.00030
Total (USD)		~\$0.00042 / message

Line item	Calculation	Cost
Total (BDT)		~৳0.046 / message

7. Symptom Checker Agent

File	backend/src/agents/symptomAgent.js
Model	GPT-4o-mini
maxTokens	1,024
Tools	rag_search , get_patient_profile
Session	Up to 8 turns

Per-turn profile:

	Input	Output
Per turn	~1,500 tokens (system + profile + RAG + history)	~400 tokens
Per session (8 turns)	~12,000 tokens	~3,200 tokens

Line item	Calculation	Cost
Input (session)	$(12,000 \div 1M) \times \0.15	\$0.00180
Output (session)	$(3,200 \div 1M) \times \$0.60$	\$0.00192
Total (USD)		~\$0.0037 / session
Total (BDT)		~৳0.41 / session

8. Symptom Summarization

File	backend/src/utils/llmUtils.js
Model	GPT-4o-mini
Prompt	Small
Input	~300 tokens
Output	~150 tokens

Line item	Calculation	Cost
Input	$(300 \div 1M) \times \$0.15$	\$0.000045
Output	$(150 \div 1M) \times \$0.60$	\$0.000090
Total (USD)		~\$0.000135 / call
Total (BDT)		~৳0.015 / call

9. Drug Interaction Check

File	backend/src/agents/drugInteractionAgent.js
Model	GPT-4o-mini (OPENAI_AGENT_MODEL)
maxTokens	3,000
Tools	check_local_db , rxnorm_lookup , medscape_lookup , web_search
Turns	~3–4 tool rounds

	Input	Output
Per turn	~2,000 tokens	~1,000 tokens
Per check (4 turns)	~8,000 tokens	~4,000 tokens

Line item	Calculation	Cost
Input	$(8,000 \div 1M) \times \$0.15$	\$0.00120
Output	$(4,000 \div 1M) \times \$0.60$	\$0.00240
Total (USD)		~\$0.0036 / check
Total (BDT)		~৳0.40 / check

10. Health Insights Generation

File	backend/src/agents/insightsAgent.js (generateInsights)
Model	GPT-4o
maxTokens	2,048
Tool rounds	~3

	Input	Output
Per round	~3,000 tokens	~2,048 tokens
Full call (3 rounds)	~9,000 tokens	~6,144 tokens

Line item	Calculation	Cost
Input	$(9,000 \div 1M) \times \$2.50$	\$0.02250
Output	$(6,144 \div 1M) \times \$10.00$	\$0.06144
Total (USD)		~\$0.084 / generation
Total (BDT)		~৳9.24 / generation

💡 Insights are **cached for 2 hours** on the frontend — not incurred on every page load.

11. Doctor Summary — AI-Generated Visit Brief

File	<code>backend/src/agents/insightsAgent.js (generateDoctorSummary)</code>
Model	Claude Sonnet 4.6 (direct Anthropic fetch)
max_tokens	900
Input	~1,500 tokens (patient profile, conditions, meds, allergies)
Output	~900 tokens

Line item	Calculation	Cost
Input	$(1,500 \div 1M) \times \$3.00$	\$0.00450
Output	$(900 \div 1M) \times \$15.00$	\$0.01350
Total (USD)		~\$0.018 / generation
Total (BDT)		~৳1.98 / generation

12. Predictive Health Analysis

File	<code>backend/src/agents/predictiveAgent.js</code>
Model	GPT-4o-mini (<code>OPENAI_PREDICTIVE_MODEL</code>)
Input	~2,000 tokens (vitals + lab trends + patient profile)

Output	~1,500 tokens (risk predictions + recommendations)
--------	--

Line item	Calculation	Cost
Input	$(2,000 \div 1M) \times \$0.15$	\$0.00030
Output	$(1,500 \div 1M) \times \$0.60$	\$0.00090
Total (USD)		~\$0.0012 / analysis
Total (BDT)		~৳0.13 / analysis

13. Specialty Inference — NearMe / Find Doctors

File	backend/src/controllers/placesController.js
Model	GPT-4o-mini
max_tokens	80
Input	~200 tokens (symptom list → specialty mapping)
Output	~80 tokens

Line item	Calculation	Cost
Input	$(200 \div 1M) \times \$0.15$	\$0.000030
Output	$(80 \div 1M) \times \$0.60$	\$0.000048
Total (USD)		~\$0.000078 / call
Total (BDT)		~৳0.0086 / call

14. RAG Embeddings — Vector Search

File	backend/src/rag/embeddings.js
Model	text-embedding-3-small (\$0.02 / 1M tokens)

Use case	Tokens	Cost (USD)	Cost (BDT)
Per query (symptom / report search)	~200	\$0.000004	৳0.0004
Per document chunk (ingestion)	~500	\$0.000010	৳0.0011
Full knowledge base re-embed (~10K chunks)	~5,000,000	\$0.10	৳11.00

Full Patient Journey — Estimated Total

A typical session: one prescription scan + one report upload + AI health insight.

Step	USD	BDT
Prescription scan (PrescriptoAI)	\$0.120	₹13.20
GPT-4.1 dosage pass	\$0.008	₹0.92
Lab report analysis	\$0.027	₹3.00
Symptom check session (8 turns)	\$0.004	₹0.41
Drug interaction check	\$0.004	₹0.40
Health insights generation	\$0.084	₹9.24
Doctor summary	\$0.018	₹1.98
RAG queries (~10 total)	~\$0.00004	₹0.004
Total	~\$0.265	~₹29.17

Report + prescription chat Q&A adds ~₹0.05 per message on top.